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Test Name:	**Mock Test**
Taken On:	9 Oct 2024 19:10:40 IST
Time Taken:	9 min 15 sec/ 22 min
Invited by:	Ankush
Invited on:	9 Oct 2024 19:10:29 IST
Skills Score:	
Tags Score:	Algorithms 105/105 Core CS 105/105 Easy 105/105 Problem Solving 105/105 Strings 105/105 problem-solving 105/105

100%
105/105
scored in **Mock Test** in 9 min 15 sec on 9 Oct 2024 19:10:40 IST

Recruiter/Team Comments:

No Comments.

	Question Description	Time Taken	Score	Status
Q1	Palindrome Index > **Coding**	9 min 2 sec	105/ 105	✓

QUESTION 1

✓ Correct Answer

Score 105

Palindrome Index > Coding

Strings | Algorithms | Easy | problem-solving | Core CS | Problem Solving

QUESTION DESCRIPTION

Given a string of lowercase letters in the range ascii[a-z], determine the index of a character that can be removed to make the string a palindrome. There may be more than one solution, but any will do. If the word is already a palindrome or there is no solution, return *-1*. Otherwise, return the index of a character to remove.

Example

$s = \text{"bcbc"}$

Either remove *'b'* at index 0 or *'c'* at index 3.

Function Description

Complete the *palindromeIndex* function in the editor below.

palindromeIndex has the following parameter(s):

- *string s:* a string to analyze

Returns

- *int:* the index of the character to remove or -1

Input Format

The first line contains an integer q, the number of queries.
Each of the next q lines contains a query string s.

Constraints

- $1 \leq q \leq 20$
- $1 \leq \text{length of } s \leq 10^5 + 5$
- All characters are in the range ascii[a-z].

Sample Input

```
STDIN   Function
-----   --------
3       q = 3
aaab    s = 'aaab' (first query)
baa     s = 'baa'  (second query)
aaa     s = 'aaa'  (third query)
```

Sample Output

```
3
0
-1
```

Explanation

Query 1: "aaab"
Removing *'b'* at index 3 results in a palindrome, so return 3.

Query 2: "baa"
Removing *'b'* at index 0 results in a palindrome, so return 0.

Query 3: "aaa"
This string is already a palindrome, so return -1. Removing any one of the characters would result in a palindrome, but this test comes first.

Note: The custom checker logic for this challenge is available here.

CANDIDATE ANSWER

Language used: **Python 3**

```

#
# Complete the 'palindromeIndex' function below.
#
# The function is expected to return an INTEGER.
# The function accepts STRING s as parameter.
#

def isPalindrome(s, b, e):
    i = b
    j = e

    while i <= j and s[i] == s[j]:
        i += 1
        j -= 1
```

```python

    return i > j



def palindromeIndex(s):
    b = 0
    e = len(s) - 1

    while b <= e and s[b] == s[e]:
        b += 1
        e -= 1

    if b > e:
        return -1

    lp = isPalindrome(s, b, e-1)
    rp = isPalindrome(s, b+1, e)

    if (lp is False and rp is False):
        return -1

    if lp is False:
        return b

    return e

```

TESTCASE	DIFFICULTY	TYPE	STATUS	SCORE	TIME TAKEN	MEMORY USED
Testcase 1	Easy	Sample case	Success	0	0.0828 sec	14.6 KB
Testcase 2	Medium	Hidden case	Success	5	0.0815 sec	14.6 KB
Testcase 3	Medium	Hidden case	Success	5	0.0895 sec	14.7 KB
Testcase 4	Medium	Hidden case	Success	5	0.0862 sec	14.6 KB
Testcase 5	Medium	Hidden case	Success	5	0.0941 sec	14.6 KB
Testcase 6	Medium	Hidden case	Success	5	0.1061 sec	14.6 KB
Testcase 7	Medium	Hidden case	Success	5	0.101 sec	14.6 KB
Testcase 8	Medium	Hidden case	Success	5	0.106 sec	14.6 KB
Testcase 9	Hard	Hidden case	Success	10	0.1255 sec	14.6 KB
Testcase 10	Hard	Hidden case	Success	10	0.1078 sec	14.6 KB
Testcase 11	Hard	Hidden case	Success	10	0.0952 sec	14.6 KB
Testcase 12	Hard	Hidden case	Success	10	0.1319 sec	14.6 KB
Testcase 13	Hard	Hidden case	Success	10	0.0964 sec	14.6 KB
Testcase 14	Hard	Hidden case	Success	10	0.103 sec	14.6 KB
Testcase 15	Hard	Hidden case	Success	10	0.1293 sec	14.6 KB

No Comments